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Research paper

Scaled Brownian motion with random anomalous diffusion exponent

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ABSTRACT

The scaled Brownian motion (SBM) is regarded as one of the paradigmatic random processes, characterized by anomalous diffusion through the diffusion exponent. It is a Gaussian, self-similar process with independent increments and has found applications across various fields, from turbulence and stochastic hydrology to biophysics. In our paper, inspired by recent single particle tracking biological experiments, we introduce a process called scaled Brownian motion with random exponent (SBMRE), which retains the key features of SBM at the level of individual trajectories, but with anomalous diffusion exponents that vary randomly across trajectories.

We discuss the main probabilistic properties of SBMRE, including its probability density function (pdf) and the qth absolute moment. Additionally, we present the expected value of the time-averaged mean squared displacement (TAMSD) and the ergodicity breaking parameter. Furthermore, we analyze the pdf of the first hitting time in a semi-infinite domain, the martingale property of SBMRE, and its stochastic exponential. As special cases, we consider two distributions for the anomalous diffusion exponent, namely the mixture of two point distributions and beta distribution, and explore the asymptotics of the corresponding characteristics. Theoretical results for SBMRE are validated through numerical simulations and compared with the analogous characteristics of SBM.

1. Introduction

Stochastic processes exhibiting anomalous diffusion behavior are ubiquitous across many fields. They are characterized by a non-linear, often power-law, second moment, specifically $\mathbb{E}\left[X^2(t)\right] \sim t^{\mu}$ [1]. The parameter μ is referred to as the anomalous diffusion exponent. Depending on the value of μ, one can distinguish between sub-diffusive ($\mu < 1$) and super-diffusive ($\mu > 1$) behavior. When $\mu = 1$, the process exhibits a linear time dependence, which is the hallmark of normal (ordinary) Brownian diffusion [2].

Mathematically, ordinary Brownian motion, or the Wiener process, is a Gaussian, self-similar process with stationary independent increments [3]. There are two generic Gaussian processes that exhibit anomalous diffusion behavior. A classic example is fractional Brownian motion (FBM), introduced by Kolmogorov [4] and further studied by Mandelbrot and van Ness [5], particularly in the context of economic time series analysis. FBM is characterized by the Hurst exponent $H \in (0,1)$, which governs its anomalous diffusion behavior as $\mu = 2H$. It is the only Gaussian self-similar process with stationary, power-law correlated increments, closely associated with long-range phenomena [6]. FBM has found numerous applications in various fields, including hydrology [7,8],

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telecommunications and signal processing [9–11], image analysis [12,13], economics [14–16], and biological systems, such as single particle tracking experiments [17–21], among others.

Another Gaussian process exhibiting anomalous diffusion behavior is scaled Brownian motion (SBM), which was implicitly introduced in the context of turbulent diffusion in [22] and later rediscovered in [23] as a simple time-changed subordination of Brownian motion by a deterministic power function. This process shares characteristics with FBM, namely it is Gaussian and self-similar. However, unlike FBM, its increments are independent (which implies Markovianity) and non-stationary.

Many authors have examined SBM from a theoretical perspective. For instance, [24,25] discuss the asymptotics of the supremum of SBM, while [26] delves into the Lamperti transformation for SBM. In [27], SBM is considered as the mean-field model for continuous-time random walks. [28] demonstrates the highly non-stationary behavior of SBM in a confining potential, while the authors of [27–29] analyze ergodicity breaking in SBM. [30] explores the power spectral density of SBM. The literature also addresses various modifications of SBM. For example, [31,32] examine ultraslow and underdamped versions of SBM, respectively, while [33,34] discuss SBM with resetting. [35] proposes a modification of SBM to make it stationary. Further studies include SBM with a random diffusion coefficient [36], SBM in a quenched disorder environment [37], and scaled geometric Brownian motion [38]. Recently, [39] explored FBM and SBM on a sphere, investigating the effects of long-time correlations on navigation strategies. SBM has also found intriguing applications, particularly in the physical sciences and biology, as evidenced by works such as [28,31,40–42]. Interested readers can also explore [43–45] for different approaches proposed for testing SBM.

With recent developments in single particle tracking techniques, it has become clear that "standard" anomalous diffusion models, such as FBM and SBM, may be inadequate for describing certain complex systems. For example, modern experiments suggest that the motion of biological cells exhibits anomalous diffusion behavior at the level of individual trajectories. However, the anomalous diffusion exponent varies from one trajectory to another (see, e.g., [46–49]). In such cases, a natural generalization of the standard anomalous diffusion model is to account for the randomness of parameters characterizing the random process, such as the diffusivity and/or the anomalous diffusion exponent. This idea of doubly stochastic behavior is central to the superstatistics approach [50,51], which has recently been further developed in the concept of 'diffusing diffusivity' proposed in [52] and explored in a series of papers (see, e.g., the review [53] and references therein).

The multifractional generalization of FBM, which allows the Hurst exponent to be a stationary random process, has been discussed in the mathematical literature [54,55]. A simplified version, called FBMRE, which considers the Hurst exponent as a variable that randomly changes from trajectory to trajectory, is analyzed in [56], where interesting effects such as accelerating diffusion and persistence transitions over time are found. We also refer the reader to recent publications that discuss FBM-based models with various scenarios for random parameters [57–60].

In this paper, we investigate scaled Brownian motion with a random exponent (SBMRE), which was introduced in [61]. In that work, the authors analyzed the probability density function (pdf) and the second moment of SBMRE. In contrast to [61], where all results are presented in integral form, our approach is based on the moment-generating function. Apart from analyzing the pdf and the second moment of SBMRE, we extend the mathematical description by presenting the transition pdf and adding the q-th absolute moment of SBMRE, the expected value of the time-averaged mean squared displacement (TAMSD), the ergodicity breaking parameter, and the first hitting time pdf. Finally, we discuss the martingale property of SBMRE and its stochastic exponential. As a special case, we examine two general distributions of the anomalous diffusion exponent: namely, the mixture of two point distributions and beta distribution. In these cases, we present the aforementioned characteristics of SBMRE and analyze their asymptotics. Our theoretical results are supported by numerical analysis that demonstrates the specific behavior of the considered stochastic process and its differences in relation to the classical SBM.

The rest of the paper is organized as follows. In Section 2, we recall all the necessary definitions. In Section 3, we remind readers of the important properties of SBM, while in Section 4, we present the same characteristics for SBMRE concerning the general distribution of the anomalous diffusion exponent. In Section 5, we provide results for two exemplary distributions of the anomalous diffusion exponent. In Section 6, we validate the theoretical results with numerical simulations. The final section concludes the paper.

2. Preliminaries

In this section, we introduce all definitions and notations used in the following sections. We assume that all processes take values in $\mathbb{R}$. If we do not express it explicitly, we assume, that corresponding probability space is $\left(\Omega, \mathcal{F}, \mathbb{P}, \left\{\mathcal{F}_t\right\}\right)$.

Definition 2.1. Let $\{X(t)\}_{t\geq 0}$ be the stochastic process . Then, the time averaged mean square displacement (TAMSD) of the process $\{X(t)\}_{t\geq 0}$ is defined as follows [19]:

$$\delta(\tau) = \frac{1}{T-\tau}\int_0^{T-\tau} (X(t+\tau) - X(t))^2\, dt, \tag{1}$$

where $\tau \in [0, T)$ is the time lag of measured time series (the width of a sliding window), and $T \in (0, \infty)$ is a time horizon (the trajectory length).

Apparently, the TAMSD is a random process. In order to quantify its randomness, the normalized variance called ergodicity breaking parameter is introduced.

Definition 2.2. Let $\{X(t)\}_{t\geq 0}$ be the stochastic process. Then, the ergodicity breaking parameter of the process $\{X(t)\}_{t\geq 0}$ is defined as follows [62,63]:

$$EB(\tau) = \frac{\mathbb{E}\left[\delta^2(\tau)\right] - (\mathbb{E}[\delta(\tau)])^2}{(\mathbb{E}[\delta(\tau)])^2} = \frac{Var(\delta(\tau))}{\mathbb{E}[\delta(\tau)]^2} = \frac{\mathcal{N}(\tau)}{D(\tau)}. \tag{2}$$

Next, we define the hitting time in the barrier $b > 0$ of a given stochastic process.

Definition 2.3. Let $\{X(t)\}_{t\geq 0}$ be the stochastic process. Then the hitting time in the barrier $b > 0$ of the process $\{X(t)\}_{t\geq 0}$ is a random variable defined as follows:

$$\tau_b = \inf\{t \geq 0 : X(t) = b\}. \tag{3}$$

The next two definitions are the quadratic variation and the stochastic exponent. These are important tools when we focus on the martingale properties of the process.

Definition 2.4. Let $\{X(t)\}_{t\geq 0}$ be the semimartingale (for more details, see [64]). Then the quadratic variation of the process $\{X(t)\}_{t\geq 0}$ is defined by [64]:

$$[X]_t = \lim_{\Delta t_k \to 0} \sum_{t_k \leq t} \left(X\left(t_{k+1}\right) - X\left(t_k\right)\right)^2, \tag{4}$$

where $0 = t_1 < t_2 < \cdots < t_n = t$, $\Delta t_k = t_{k+1} - t_k$ and limit is taken in probability.

Definition 2.5. Let $\{X(t)\}_{t\geq 0}$ be the semimartingale. Then the stochastic exponential of the process $\{X(t)\}_{t\geq 0}$ is given by [64]:

$$Y(t) = \exp\left\{\lambda X(t) - \frac{\lambda^2}{2}[X]_t\right\}, \ \lambda > 0. \tag{5}$$

The last two definitions are the big $\mathcal{O}$ and the little o notation. They are important for providing asymptotic formulas. In both definitions, we consider functions $f(\cdot)$ and $g(\cdot)$ whose domain and co-domain are $\mathbb{R}$.

Definition 2.6. We say, that $f(x) \in \mathcal{O}(g(x))$ as $x \to \infty$, if there exist $c > 0$ and $x_0 \in \mathbb{R}$, such that for all $x \geq x_0$ we have [65]:

$$|f(x)| \leq c \cdot g(x). \tag{6}$$

Respectively, we say that $f(x) \in o(g(x))$ as $x \to \infty$, if relation (6) holds for all $c > 0$.

3. Scaled Brownian motion

In this section we recall the definition of the scaled Brownian motion and describe its main properties.

Definition 3.1. Let $\{B(t)\}_{t\geq 0}$ be the standard Brownian motion. Then, the scaled Brownian motion (SBM) is defined as follows:

$$B_\alpha(t) = B(t^\alpha), \tag{7}$$

where $\alpha > 0$ is an anomalous diffusion exponent. It is a Gaussian self-similar process with independent increments (Markovianity), zero mean, and autocovariance $Cov\left(B_\alpha(s), B_\alpha(t)\right) = \min\{t^\alpha, s^\alpha\}$. SBM is subdiffusive process for $\alpha < 1$ and superdiffusive process for $\alpha > 1$. For $\alpha = 1$ the SBM becomes an ordinary Brownian motion. The transition probability density function is given by:

$$p\left(x, t \mid x_0, t_0\right) = \frac{1}{\sqrt{2\pi\left(t^\alpha - t_0^\alpha\right)}} \exp\left\{-\frac{\left(x - x_0\right)^2}{2\left(t^\alpha - t_0^\alpha\right)}\right\}, \quad x, x_0 \in \mathbb{R}. \tag{8}$$

The SBM can also be represented by the following stochastic differential equation (SDE) [23]:

$$dB_\alpha(t) = \sqrt{\alpha} t^{\frac{\alpha-1}{2}} dB(t), \quad B_\alpha(0) = 0. \tag{9}$$

The solution of (9) reads

$$B_\alpha(t) = \int_0^t \sqrt{\alpha} s^{\frac{\alpha-1}{2}} dB(s). \tag{10}$$

In the next part of this section we present the main properties of SBM.

Fact 3.1. *Let $\left\{B_\alpha(t)\right\}_{t\geq 0}$ be the SBM. Then, the expected value of the TAMSD for SBM is given by [28]:*

$$\mathbb{E}\left[\delta_\alpha(\tau)\right] = \frac{T^{\alpha+1} - \tau^{\alpha+1} - (T-\tau)^{\alpha+1}}{(\alpha+1)(T-\tau)}. \tag{11}$$

For $\tau/T \ll 1$ the expected value of the TAMSD for SBM (11) asymptotically behaves like [29]:

$$\mathbb{E}\left[\delta_\alpha(\tau)\right] \sim \tau/T^{1-\alpha}. \tag{12}$$

Fact 3.2. *Let $\left\{B_\alpha(t)\right\}_{t\geq 0}$ be the SBM. Then, the ergodicity breaking parameter for SBM is given by [29]*

$$EB_\alpha(\tau) = \frac{\mathcal{N}_\alpha(\tau)}{D_\alpha(\tau)}, \tag{13}$$

where

$$\begin{aligned}\mathcal{N}_\alpha(\tau) = \frac{4\tau^{2\alpha+2}}{(T-\tau)^2}\Bigg[&\frac{(T/\tau-1)^{2\alpha+1}}{2\alpha+1} + \frac{(3\alpha+1)(T/\tau-1)^{2\alpha+2}}{2(\alpha+1)^2(2\alpha+1)} - \frac{2(T/\tau)^{\alpha+1}(T/\tau-1)^{\alpha+1}}{(\alpha+1)^2} + \\ &+ \frac{(T/\tau)^{2\alpha+2}}{2(\alpha+1)(2\alpha+1)} - \frac{\left(2\alpha^2+\alpha+1\right)}{2(\alpha+1)^2(2\alpha+1)} + \frac{2}{\alpha+1}\int_0^{T/\tau-1} x^{\alpha+1}(x+1)^\alpha\,dx\Bigg]\end{aligned} \tag{14}$$

and $D_\alpha(\tau)$ is the square of (11).

For $\tau/T \ll 1$ the EB parameter for SBM (13) asymptotically behaves like [29]:

$$EB_\alpha(\tau) \sim \begin{cases} C(\alpha)\left(\frac{\tau}{T}\right)^{2\alpha}; & 0<\alpha<1/2 \\ \frac{\tau}{12T}\left[\log\left(\frac{T}{\tau}\right)+2\log(2)-\frac{5}{6}\right]; & \alpha = 1/2 \\ \frac{1}{3}\frac{\alpha^2}{2\alpha-1}\frac{\tau}{T}; & \alpha > 1/2, \end{cases} \tag{15}$$

where $C(\alpha) = \frac{(1-\alpha)(2-\alpha)\mathbb{B}(\alpha+2,1-2\alpha)-2\left(\alpha^2+\alpha-1\right)}{2(\alpha+1)^2(\alpha+1)}$.

Fact 3.3. *Let $\left\{B_\alpha(t)\right\}_{t\geq 0}$ be the SBM. Let $\tau_b = \inf\left\{t\geq 0 : B_\alpha(t) = b\right\}$ be the time of first hitting a point $b>0$. Then, the pdf of τ_b is given by [33]*

$$f_{\tau_b}(t) = \frac{\alpha b}{\sqrt{2\pi}} e^{-\frac{b^2}{2t^\alpha}} t^{-1-\frac{\alpha}{2}}, \quad t\geq 0. \tag{16}$$

Theorem 3.1. *Let $\left\{B_\alpha(t)\right\}_{t\geq 0}$ be the SBM. Then, SBM is a martingale with respect to its natural filtration. Its quadratic variation $\left[B_\alpha\right]_t = t^\alpha$. The stochastic exponential of SBM is also a martingale with respect to its natural filtration.*

Proof. We start with the proof that SBM is a martingale. The first moment of absolute value of SBM is, obviously, finite. For $s<t$ we have:

$$\mathbb{E}\left[B_\alpha(t) \mid \mathcal{F}_s\right] = \mathbb{E}\left[B_\alpha(t) - B_\alpha(s) + B_\alpha(s) \mid \mathcal{F}_s\right] = \mathbb{E}\left[B_\alpha(t) - B_\alpha(s)\right] + B_\alpha(s) = B_\alpha(s). \tag{17}$$

Next, using the SDE representation (9), we calculate the quadratic variation of the SBM. Namely, we have:

$$\left[B_\alpha\right]_t = \int_0^t \left(\sqrt{\alpha}s^{\frac{\alpha-1}{2}}\right)^2 ds = \int_0^t \alpha s^{\alpha-1} ds = t^\alpha. \tag{18}$$

Now, we will show that the stochastic exponential of SBM is a martingale. Indeed, we have:

$$\mathbb{E}\left[e^{B_\alpha(t)} \mid \mathcal{F}_s\right] = \mathbb{E}\left[e^{B_\alpha(t)-B_\alpha(s)}e^{B_\alpha(s)} \mid \mathcal{F}_s\right] = e^{B_\alpha(s)}\mathbb{E}\left[e^{B_\alpha(t)-B_\alpha(s)}\right] = e^{\frac{t^\alpha-s^\alpha}{2}}e^{B_\alpha(s)}. \tag{19}$$

The last equality is a straightforward consequence of the Gaussianity. Finally, we have the following:

$$\mathbb{E}\left[e^{B_\alpha(t)-\frac{t^\alpha}{2}} \mid \mathcal{F}_s\right] = e^{B_\alpha(s)-\frac{s^\alpha}{2}}. \tag{20}$$

As a consequence, it also gives us the integrability of stochastic exponential. □

4. Scaled Brownian motion with random anomalous diffusion exponent

In this section, we define scaled Brownian motion with random anomalous diffusion exponent and investigate its main properties.

Definition 4.1. Let $\mathcal{A}$ be some positive random variable with values from the interval $(0,K)$ with pdf $f_{\mathcal{A}}(\cdot)$ and a moment generating function $M_{\mathcal{A}}(\cdot)$:

$$M_{\mathcal{A}}(s) = \mathbb{E}\left[e^{s\mathcal{A}}\right] = \int_0^K e^{sa} f_{\mathcal{A}}(a)\,da. \tag{21}$$

$K>0$ is finite constant.

Then, the scaled Brownian motion with random anomalous diffusion exponent (SBMRE) is defined as follows:

$$B_{\mathcal{A}}(t) = B\left(t^{\mathcal{A}}\right). \tag{22}$$

We assume that $\{B(t)\}_{t\geq 0}$ and $\mathcal{A}$ are independent. SBMRE can also be represented via the following SDE:

$$dB_{\mathcal{A}}(t) = \sqrt{\mathcal{A}}t^{\frac{\mathcal{A}-1}{2}}dB(t), \quad B_{\mathcal{A}}(0) = 0. \tag{23}$$

The solution of (23) is given by:

$$B_{\mathcal{A}}(t) = \int_0^t \sqrt{\mathcal{A}}s^{\frac{\mathcal{A}-1}{2}}dB(s). \tag{24}$$

The representation via SDE (23) is a direct consequence of random time change for Brownian motion (for the details see Chapter 8 in [64]).

Fact 4.1. *Let $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ be the SBMRE. Then, its transition pdf is given by the following formula:*

$$p_{B_{\mathcal{A}}}(x,t \mid x_0,t_0) = \int_0^K \frac{1}{\sqrt{2\pi\left(t^a - t_0^a\right)}} exp\left\{\frac{-\left(x-x_0\right)^2}{2\left(t^a - t_0^a\right)}\right\} f_{\mathcal{A}}(a)\,da. \tag{25}$$

Proof. It is a direct consequence of the law of total probability. □

Fact 4.2. *Let $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ be the SBMRE. Then, the moment of order q of the absolute value of $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ is given by:*

$$\mathbb{E}\left[\left|B_{\mathcal{A}}(t)\right|^q\right] = c_q M_{\mathcal{A}}\left(\frac{1}{2}q\log t\right), \tag{26}$$

where $c_q = \frac{2^{q/2}\Gamma\left(\frac{q+1}{2}\right)}{\sqrt{\pi}}$.

Proof.

$$\begin{aligned}\mathbb{E}\left[\left|B_{\mathcal{A}}(t)\right|^q\right] &= \mathbb{E}\left[\mathbb{E}\left[\left|B_{\mathcal{A}}(t)\right|^q \mid \mathcal{A}\right]\right] = \mathbb{E}\left[\mathbb{E}\left[\left|B\left(t^{\mathcal{A}}\right)\right|^q \mid \mathcal{A}\right]\right] = \mathbb{E}\left[\mathbb{E}\left[t^{\frac{1}{2}\mathcal{A}q}\left|B(1)\right|^q \mid \mathcal{A}\right]\right] = \mathbb{E}\left[t^{\frac{1}{2}\mathcal{A}q}Z^q\right] = \\ &= c_q\mathbb{E}\left[e^{\frac{1}{2}q\mathcal{A}\log t}\right] = c_q M_{\mathcal{A}}\left(\frac{1}{2}q\log t\right),\end{aligned} \tag{27}$$

where $Z = |B(1)|^q$ and $\Gamma(\cdot)$ is the gamma function. □

Corollary 4.1. *The second moment of $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ is given by:*

$$\mathbb{E}\left[B_{\mathcal{A}}^2(t)\right] = M_{\mathcal{A}}(\log t). \tag{28}$$

Fact 4.3. *Let $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ be the SBMRE. Then, the autocovariance function of SBMRE is as follows:*

$$Cov\left(B_{\mathcal{A}}(s), B_{\mathcal{A}}(t)\right) = M_{\mathcal{A}}(\log(\min\{s,t\})). \tag{29}$$

Proof.

$$\begin{aligned}Cov\left(B_{\mathcal{A}}(s), B_{\mathcal{A}}(t)\right) &= \mathbb{E}\left[B_{\mathcal{A}}(s)B_{\mathcal{A}}(t)\right] = \mathbb{E}\left[\mathbb{E}\left[B_{\mathcal{A}}(s)B_{\mathcal{A}}(t) \mid \mathcal{A}\right]\right] = \mathbb{E}\left[\min\left\{s^{\mathcal{A}}, t^{\mathcal{A}}\right\}\right] = \\ &= \mathbb{E}\left[\min\{s,t\}^{\mathcal{A}}\right] = \mathbb{E}\left[\exp\{\mathcal{A}\log(\min\{s,t\})\right] = M_{\mathcal{A}}(\log(\min\{s,t\})).\end{aligned} \tag{30}$$

□

Fact 4.4. *Let $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ be the SBMRE. Then, the expected value of TAMSD for SBMRE is given by:*

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] = \frac{1}{T-\tau}\int_0^{T-\tau}\left(M_{\mathcal{A}}(\log(t+\tau)) - M_{\mathcal{A}}(\log t)\right)dt. \tag{31}$$

Proof.

$$\begin{aligned}\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] &= \mathbb{E}\left[\frac{1}{T-\tau}\int_0^{T-\tau}\left(B_{\mathcal{A}}(t+\tau) - B_{\mathcal{A}}(t)\right)^2 dt\right] = \mathbb{E}\left[\frac{1}{T-\tau}\int_0^{T-\tau}(B_{\mathcal{A}}^2(t+\tau) - 2B_{\mathcal{A}}(t+\tau)B_{\mathcal{A}}(t) + B_{\mathcal{A}}^2(t))dt\right] = \\ &= \frac{1}{T-\tau}\int_0^{T-\tau}\left(M_{\mathcal{A}}(\log(t+\tau)) - M_{\mathcal{A}}(\log t)\right)dt.\end{aligned} \tag{32}$$

The change in the order of integration is made according to the Fubini's theorem [66]. □

Fact 4.5. *Let $\{B_{\mathcal{A}}(t)\}_{t\geq 0}$ be the SBMRE. Then, the ergodicity breaking parameter of SBMRE is given by:*

$$EB_{\mathcal{A}}(\tau) = \frac{\mathcal{N}_{\mathcal{A}}(\tau)}{\mathcal{D}_{\mathcal{A}}(\tau)}, \tag{33}$$

where

$$\begin{aligned}\mathcal{N}_{\mathcal{A}}(\tau) = Var\left(\delta_{\mathcal{A}}(\tau)\right) = \int_0^K \frac{4\tau^{2a+2}}{(T-\tau)^2}\Bigg[&\frac{(T/\tau-1)^{2a+1}}{2a+1} + \frac{(3a+1)(T/\tau-1)^{2a+2}}{2(a+1)^2(2a+1)} - \frac{2(T/\tau)^{a+1}(T/\tau-1)^{a+1}}{(a+1)^2} + \\ &+ \frac{(T/\tau)^{2a+2}}{2(a+1)(2a+1)} - \frac{\left(2a^2+a+1\right)}{2(a+1)^2(2a+1)} + \frac{2}{a+1}\int_0^{T/\tau-1}x^{a+1}(x+1)^a dx\Bigg] f_{\mathcal{A}}(a)\,da - \\ &- \int_0^K\left(\frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)}\right)^2 f_{\mathcal{A}}(a)\,da + \left(\int_0^K \frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)} f_{\mathcal{A}}(a)\,da\right)^2\end{aligned} \tag{34}$$

and

$$\mathcal{D}_{\mathcal{A}}(\tau)=\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right]^2=\frac{1}{(T-\tau)^2}\int_0^{T-\tau}\int_0^{T-\tau}\left(M_{\mathcal{A}}\left(\log\left(t_1+\tau\right)\right)-M_{\mathcal{A}}\left(\log t_1\right)\right)\left(M_{\mathcal{A}}\left(\log\left(t_2+\tau\right)\right)-M_{\mathcal{A}}\left(\log t_2\right)\right)dt_1dt_2. \tag{35}$$

Proof. We can rewrite the EB parameter in the following form using the law of total variance [67]:

$$EB_{\mathcal{A}}(\tau)=\frac{Var\left(\delta_{\mathcal{A}}(\tau)\right)}{\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right]^2}=\frac{\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]+Var\left(\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right)}{\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right]^2}. \tag{36}$$

To calculate the first term in the numerator we use (14) and the law of total expectation,

$$\begin{aligned}\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]=\int_0^K\frac{4\tau^{2a+2}}{(T-\tau)^2}\Bigg[&\frac{(T/\tau-1)^{2a+1}}{2a+1}+\frac{(3a+1)(T/\tau-1)^{2a+2}}{2(a+1)^2(2a+1)}-\frac{2(T/\tau)^{a+1}(T/\tau-1)^{a+1}}{(a+1)^2}+\\&+\frac{(T/\tau)^{2a+2}}{2(a+1)(2a+1)}-\frac{\left(2a^2+a+1\right)}{2(a+1)^2(2a+1)}+\frac{2}{a+1}\int_0^{T/\tau-1}x^{a+1}(x+1)^a\,dx\Bigg]f_{\mathcal{A}}(a)\,da.\end{aligned} \tag{37}$$

To calculate the second term in the numerator we use (11) and the law of total expectation,

$$\begin{aligned}Var\left(\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right)=\mathbb{E}\left[\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]^2\right]-\mathbb{E}\left[\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right]^2=\int_0^K\left(\frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)}\right)^2f_{\mathcal{A}}(a)\,da-\\-\left(\int_0^K\frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)}f_{\mathcal{A}}(a)\,da\right)^2.\end{aligned} \tag{38}$$

Finally, we obtain (34).

By taking a square of (31) and changing the square of the integral into a double integral, we get

$$\mathcal{D}_{\mathcal{A}}(\tau)=\frac{1}{(T-\tau)^2}\int_0^{T-\tau}\int_0^{T-\tau}\left(M_{\mathcal{A}}\left(\log\left(t_1+\tau\right)\right)-M_{\mathcal{A}}\left(\log t_1\right)\right)\left(M_{\mathcal{A}}\left(\log\left(t_2+\tau\right)\right)-M_{\mathcal{A}}\left(\log t_2\right)\right)dt_1dt_2. \quad\square \tag{39}$$

We observe, that if $\mathcal{A}$ is constant, then (33) reduces to (13).

Fact 4.6. *Let $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ be the SBMRE. Then, the distribution of the first hitting time in barrier b for SBMRE $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ has the following pdf:*

$$f_{\tau_b}(t)=\int_0^K\frac{ab}{\sqrt{2\pi}}e^{-\frac{b^2}{2t^a}}t^{-1-\frac{a}{2}}f_{\mathcal{A}}(a)\,da. \tag{40}$$

Proof. We use (16) and the law of total probability. $\square$

Next we discuss the martingale properties or SBMRE. In the proof, we follow the same steps as in the proof of Theorem 2.1 in [68]. First, we define the filtration $\left\{\mathcal{F}_\tau\right\}_{\tau\geq0}$ given by the following formula:

$$\mathcal{F}_\tau=\bigcap_{u>\tau}\left\{\sigma\left(B(y)\ :\ 0\leq y\leq u\right)\vee\sigma\left(y^{\mathcal{A}}\ :\ y\geq0\right)\right\}. \tag{41}$$

$\mathcal{F}_2\vee\mathcal{F}_2$ denotes the $\sigma-$ algebra generated by the union of the $\sigma-$ algebras $\mathcal{F}_1$, $\mathcal{F}_2$ and $\bigcap_{u>\tau}$ is the intersection of sets taken over all $u>\tau$.

Theorem 4.1. *Let $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ be the SBMRE. Suppose that the density of the random variable $\mathcal{A}$ is continuous. Then, $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ is a martingale with respect to filtration $\left\{\mathcal{F}_{t^{\mathcal{A}}}\right\}_{t\geq0}$, where $\left\{\mathcal{F}_\tau\right\}_{\tau\geq0}$ is given by (41). The quadratic variation of $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ is given by $\left[B_{\mathcal{A}}\right]_t=t^{\mathcal{A}}$. Furthermore, the stochastic exponential of $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ is also $\left\{\mathcal{F}_{t^{\mathcal{A}}}\right\}_{t\geq0}$-martingale.*

Proof. We start with observations that $\left\{\mathcal{F}_\tau\right\}$ is right-continuous, $B(\tau)$ is $\left\{\mathcal{F}_\tau\right\}$-martingale, and for every fixed $t_0>0$ the random variable $t_0^{\mathcal{A}}$ is the stopping time with respect to $\left\{\mathcal{F}_{t^{\mathcal{A}}}\right\}$. These facts ensure that $\mathcal{G}_t=\left\{\mathcal{F}_{t_0^{\mathcal{A}}}\right\}$ is well defined. Next, we introduce a sequence of $\left\{\mathcal{F}_\tau\right\}$-stopping times defined by

$$T_n=\inf\left\{\tau>0\ :\ |B(\tau)|=n\right\}. \tag{42}$$

Using Doob's optional sampling theorem [69] we obtain

$$\mathbb{E}\left[B\left(T_n\wedge t^{\mathcal{A}}\right)\mid\mathcal{G}_s\right]=B\left(T_n\wedge s^{\mathcal{A}}\right)\quad s<t. \tag{43}$$

Let us note that as $n\to\infty$ the left-hand side and the right-hand side of (43) converge to $\mathbb{E}\left[B\left(t^{\mathcal{A}}\right)\mid\mathcal{G}_s\right]$ and $B\left(s^{\mathcal{A}}\right)$, respectively. Thus, $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ is $\left\{\mathcal{G}_t\right\}-$ martingale.

From the SDE representation of SBMRE (23) we have that the quadratic variation of $\left\{B_{\mathcal{A}}(t)\right\}_{t\geq0}$ is given by:

$$\left[B_{\mathcal{A}}\right]_t=\int_0^t\left(\sqrt{\mathcal{A}}s^{\frac{\mathcal{A}-1}{2}}\right)^2ds=\int_0^t\mathcal{A}s^{\mathcal{A}-1}ds=t^{\mathcal{A}}. \tag{44}$$

Now we show that the stochastic exponential $\{Y(t)\}_{t\geq 0}$ of SBMRE is also a martingale. Using the Proposition 3.4 from [69] we conclude that $\{Y(t)\}_{t\geq 0}$ is a local martingale. Hence, it is sufficient to show that $\sup_{0\leq u\leq t} Y(u)$ is integrable to conclude that $\{Y(t)\}_{t\geq 0}$ is a martingale. We have the following:

$$\sup_{0\leq u\leq t} Y(u) \leq \sup_{0\leq u\leq t} \exp\left\{\lambda B\left(u^{\mathcal{A}}\right)\right\}. \tag{45}$$

In addition, $\exp\left\{\lambda B\left(u^{\mathcal{A}}\right)\right\}$ is a positive submartingale. Thus, using Doob's maximal inequality, we get:

$$\mathbb{E}\left[\left(\sup_{0\leq u\leq t} \exp\left\{\lambda B\left(u^{\mathcal{A}}\right)\right\}\right)^2\right] \leq 4\mathbb{E}\left[\exp\left\{2\lambda B\left(u^{\mathcal{A}}\right)\right\}\right]. \tag{46}$$

Now, using the properties of Gaussian random variable and the conditioning on $\sigma\left(y^{\mathcal{A}} : y\geq 0\right)$, we have:

$$\begin{aligned}\mathbb{E}\left[\exp\left\{2\lambda B\left(t^{\mathcal{A}}\right)\right\}\right] &\leq \mathbb{E}\left[\exp\left\{2\lambda^2 t^{\mathcal{A}}\right\}\right] = \sum_{n=0}^{\infty}\frac{\left(2\lambda^2\right)^n \mathbb{E}\left[t^{n\mathcal{A}}\right]}{n!} = \sum_{n=0}^{\infty}\frac{\left(2\lambda^2\right)^n \mathcal{M}_{\mathcal{A}}\left(n\log(t)\right)}{n!} = \\ &= \sum_{n=0}^{\infty}\frac{\left(2\lambda^2\right)^n \int_0^K e^{an\log(t)} f_{\mathcal{A}}(a)\,da}{n!} = \int_0^K f_{\mathcal{A}}(a)\sum_{n=0}^{\infty}\frac{\left(2\lambda^2\right)^n e^{an\log(t)}}{n!}da = \int_0^K e^{2t^a\lambda^2} f_{\mathcal{A}}(a)\,da \leq e^{2t^c\lambda^2} f_{\mathcal{A}}(c),\end{aligned} \tag{47}$$

where $c\in(0,K)$. The last inequality is a straightforward consequence of the mean value theorem and the change in order of integration and summation is performed according to the dominated convergence theorem [66]. Finally, $\mathbb{E}\left[\sup_{0\leq u\leq t} Y(u)\right] < \infty$, thus $\{Y(t)\}_{t\geq 0}$ is a martingale. □

5. Selected distributions of anomalous diffusion exponent

In this section, we apply the general formulas calculated in the previous section to characterize the SBMRE with two-point and beta distributions of random anomalous diffusion exponent. For simplicity, when we consider transition pdf, we set $x_0, t_0 = 0$ and write $p_{B_{\mathcal{A}}}\left(x,t \mid x_0,t_0\right) = p_{B_{\mathcal{A}}}(x,t)$.

Example 5.1. First, we consider the case of the simplest distribution of the random variable $\mathcal{A}$, that is the mixture of two point distributions, concentrated at two points $A_1, A_1\in(0,K)$, $A_1 < A_2$ with pdf given by

$$f_{\mathcal{A}}(a) = p\delta\left(a-A_1\right) + (1-p)\,\delta\left(a-A_2\right), \tag{48}$$

where the weight parameter $p\in(0,1)$ and $\delta(\cdot)$ is the Dirac delta. For brevity in what follows we use the name "two-point distribution", which of course should not be mistakenly thought to be a transition probability. The moment generating function of the two-point distribution reads as

$$M_{\mathcal{A}}(s) = pe^{A_1 s} + (1-p)\,e^{A_2 s}. \tag{49}$$

The pdf of SBMRE in this case is given by:

$$p_{B_{\mathcal{A}}}(x,t) = \frac{p}{\sqrt{2\pi t^{A_1}}}\exp\left\{-\frac{x^2}{2t^{A_1}}\right\} + \frac{1-p}{\sqrt{2\pi t^{A_2}}}\exp\left\{-\frac{x^2}{2t^{A_2}}\right\}, \quad x\in\mathbb{R}. \tag{50}$$

Obviously, (50) is a mixture of pdfs of two Gaussian distributed random variables with zero mean and variances equal to t^{A_1} and t^{A_2}, respectively.

The second moment of SBMRE has the form

$$\mathbb{E}\left[B^2_{\mathcal{A}}(t)\right] = pt^{A_1} + (1-p)\,t^{A_2}. \tag{51}$$

Notify that for $t\ll 1$ $t^{A_1}\in o\left(t^{A_2}\right)$ and for $t\gg 1$ $t^{A_2}\in o\left(t^{A_1}\right)$. Thus, the asymptotics behave as

$$\mathbb{E}\left[B^2_{\mathcal{A}}(t)\right] \sim t^{A}, \tag{52}$$

where $A = A_1$ for $t\to 0$ and $A = A_2$ for $t\to\infty$. This effect is called accelerating diffusion [70–72].

The expectation of TAMSD reads

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] = \frac{1}{T-\tau}\left(p\frac{T^{A_1+1}-(T-\tau)^{A_1+1}-\tau^{A_1+1}}{A_1+1} + (1-p)\frac{T^{A_2+1}-(T-\tau)^{A_2+1}-\tau^{A_2+1}}{A_2+1}\right), \tag{53}$$

Taking the first three components of Taylor expansion of $(T-\tau)^{A_1+1}$ and $(T-\tau)^{A_2+1}$ for $\tau/T\ll 1$ we obtain:

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] \sim p\left(\tau T^{A_1-1} + \frac{\tau^{A_1+1}}{\left(A_1+1\right)T} - A_1\tau^2 T^{A_1-2}\right) + (1-p)\left(\tau T^{A_2-1} + \frac{\tau^{A_2+1}}{\left(A_2+1\right)T} - A_2\tau^2 T^{A_2-2}\right). \tag{54}$$

Thus, for the asymptotics at $\tau/T\ll 1$ (the inequality typical for experiments [19]) we get:

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] \sim \frac{\tau}{T}\left(pT^{A_1} + (1-p)T^{A_2}\right), \tag{55}$$

which indicates that for the trajectories long enough the T-dependence of the expectation value is determined by larger exponent.

The EB parameter for two-point diffusion of anomalous diffusion exponent is calculated by the use of formula (36). The asymptotics for fixed value of τ and large T are as follows, see Appendix A for the details,

$$EB_{\mathcal{A}}(\tau) \sim \begin{cases} \frac{pC(A_1)\tau^{2A_1}+(1-p)C(A_2)\tau^{2A_2}+p(1-p)\left(T^{A_1}-T^{A_2}\right)^2}{\left(pT^{A_1}+(1-p)T^{A_2}\right)^2}; & A_1, A_2 < 1/2 \\ \frac{pC(A_1)\tau^{2A_1}+\frac{1-p}{12}\tau\left[\log\left(\frac{T}{\tau}\right)+2\log(2)-\frac{5}{6}\right]+p(1-p)\left(T^{A_1}-T^{1/2}\right)^2}{\left(pT^{A_1}+(1-p)T^{1/2}\right)^2}; & A_1 < 1/2,\ A_2 = 1/2 \\ \frac{pC(A_1)\tau^{2A_1}+\frac{(1-p)A_2^2}{3(2A_2-1)}\tau T^{2A_2-1}+p(1-p)\left(T^{A_1}-T^{A_2}\right)^2}{\left(pT^{A_1}+(1-p)T^{A_2}\right)^2}; & A_1 < 1/2,\ A_2 > 1/2 \\ \frac{\frac{p}{12}\tau\left[\log\left(\frac{T}{\tau}\right)+2\log(2)-\frac{5}{6}\right]+\frac{(1-p)A_2^2}{3(2A_2-1)}\tau T^{2A_2-1}+p(1-p)\left(T^{1/2}-T^{A_2}\right)^2}{\left(pT^{1/2}+(1-p)T^{A_2}\right)^2}; & A_1 = 1/2,\ A_2 > 1/2 \\ \frac{\frac{pA_1^2}{3(2A_1-1)}\tau T^{2A_1-1}+\frac{(1-p)A_2^2}{3(2A_2-1)}\tau T^{2A_2-1}+p(1-p)\left(T^{A_1}-T^{A_2}\right)^2}{\left(pT^{A_1}+(1-p)T^{A_2}\right)^2}; & A_1 > 1/2,\ A_2 > 1/2, \end{cases} \tag{56}$$

where $C(\alpha)$ is the same as in (15). Let us note that for $p = 0$ or $p = 1$ and in case $A_1 = A_2$, the formula (56) reduces to the asymptotics of EB for SBM, see (15). Note the appearance of the last term in the numerator with the prefactor $p(1-p)$, which is absent in the SBM case. From formula (56) we can observe the following universal limit of EB parameter for SBMRE with two-point distribution of anomalous diffusion exponent,

$$\lim_{T\to\infty} EB_{\mathcal{A}}(\tau) = \begin{cases} \frac{p}{1-p}, & \text{for } p \neq 0 \text{ and } p \neq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{57}$$

In contrast to the SBM case, in the limit of long trajectories the EB parameter does not depend on the sliding window width τ. This interesting property is checked by Monte Carlo simulations in the next section.

Further, the first hitting time in the barrier b in the considered case is as follows:

$$f_{\tau_b}(t) = \frac{pA_1 b}{\sqrt{2\pi}} \exp\left\{-\frac{b^2}{2t^{A_1}}\right\} t^{-1-\frac{A_1}{2}} + \frac{(1-p)A_2 b}{\sqrt{2\pi}} \exp\left\{-\frac{b^2}{2t^{A_2}}\right\} t^{-1-\frac{A_2}{2}}. \tag{58}$$

Notify that for $t \gg 1$ we have the following: $\exp\left\{-\frac{b^2}{2t^{A_1}}\right\}, \exp\left\{-\frac{b^2}{2t^{A_1}}\right\} \in \mathcal{O}(1)$ and $t^{-1-\frac{A_2}{2}} \in o\left(t^{-1-\frac{A_1}{2}}\right)$. Taking into account these facts, we obtain the following asymptotic formula for (58) in case $t \gg 1$:

$$f_{\tau_b}(t) \sim t^{-1-A_1/2}. \tag{59}$$

Example 5.2. The second considered distribution is the beta distribution on the interval $[A_1, A_2]$, $0 < A_1 < A_2$ with the pdf

$$f_{\mathcal{A}}(a) = \frac{(a-A_1)^{\gamma-1}(A_2-a)^{\beta-1}}{\mathbb{B}(\gamma,\beta)(A_2-A_1)^{\gamma+\beta-1}} \mathbb{1}_{[A_1,A_2]}, \tag{60}$$

where $\gamma, \beta > 0$ are beta distribution parameters and $\mathbb{B}(\cdot,\cdot)$ is beta function. The moment generating function of the beta distribution is given by

$$M_{\mathcal{A}}(s) = e^{A_1 s}\, {}_1F_1\left(\gamma, \gamma+\beta, s(A_2-A_1)\right), \tag{61}$$

where ${}_1F_1(\cdot,\cdot,\cdot)$ is a confluent hypergeometric function (we refer to [73] for more details).

The pdf of SBMRE in this case is given by:

$$p_{B_{\mathcal{A}}}(x,t) = \frac{1}{(A_2-A_1)^{\gamma+\beta-1}\mathbb{B}(\gamma,\beta)} \int_{A_1}^{A_2} \frac{(a-A_1)^{\gamma-1}(A_2-a)^{\beta-1}\exp\left\{-\frac{x^2}{2t^a}\right\}}{\sqrt{2\pi t^a}} da, \quad x \in \mathbb{R}. \tag{62}$$

In the case considered one can express the second moment of SBMRE as

$$\mathbb{E}\left[B_{\mathcal{A}}^2(t)\right] = t^{A_1}\, {}_1F_1\left(\gamma, \gamma+\beta, (A_2-A_1)\log(t)\right). \tag{63}$$

Using the asymptotics of the hypergeometric function [73] we obtain that (63) behaves asymptotically as

$$\mathbb{E}\left[B_{\mathcal{A}}^2(t)\right] \sim \frac{\Gamma(\gamma+\beta)t^{A_1}}{\Gamma(\gamma)\left((A_2-A_1)\log(1/t)\right)^{\gamma}} \tag{64}$$

for $t \ll 1$ and

$$\mathbb{E}\left[B_{\mathcal{A}}^{2}(t)\right] \sim \frac{\Gamma(\gamma+\beta)\, t^{A_2}}{\Gamma(\gamma)\left(\left(A_2-A_1\right)\log(t)\right)^{\beta}} \tag{65}$$

for $t \gg 1$.

The expectation of TAMSD for SBMRE with beta distributed anomalous diffusion exponent reads

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right] = \frac{1}{T-\tau}\int_0^{T-\tau}\left(e^{A_1\log(t+\tau)}\,{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log(t+\tau)\right) - e^{A_1\log t}\,{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log t\right)\right)dt. \tag{66}$$

The EB parameter in this case takes the form

$$EB_{\mathcal{A}}(\tau) = \frac{\mathcal{N}_{\mathcal{A}}(\tau)}{\mathcal{D}_{\mathcal{A}}(\tau)}, \tag{67}$$

where

$$\begin{aligned}\mathcal{N}_{\mathcal{A}}(\tau) = {}& \frac{1}{\mathbb{B}(\gamma,\beta)\left(A_2-A_1\right)^{\gamma+\beta-1}}\int_{A_1}^{A_2}\frac{4\tau^{2a+2}}{(T-\tau)^2}\left[\frac{(T/\tau-1)^{2a+1}}{2a+1}+\frac{(3a+1)(T/\tau-1)^{2a+2}}{2(a+1)^2(2a+1)}-\frac{2(T/\tau)^{a+1}(T/\tau-1)^{a+1}}{(a+1)^2}+\right.\\ &\left.+\frac{(T/\tau)^{2a+2}}{2(a+1)(2a+1)}-\frac{\left(2a^2+a+1\right)}{2(a+1)^2(2a+1)}+\frac{2}{a+1}\int_0^{T/\tau-1}x^{a+1}(x+1)^a\,dx\right]\left(a-A_1\right)^{\gamma-1}\left(A_2-a\right)^{\beta-1}da+\\ &+\frac{1}{\mathbb{B}(\gamma,\beta)\left(A_2-A_1\right)^{\gamma+\beta-1}}\int_{A_1}^{A_2}\left(\frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)}\right)^2\left(a-A_1\right)^{\gamma-1}\left(A_2-a\right)^{\beta-1}da-\\ &-\left(\frac{1}{\mathbb{B}(\gamma,\beta)\left(A_2-A_1\right)^{\gamma+\beta-1}}\int_{A_1}^{A_2}\frac{T^{a+1}-\tau^{a+1}-(T-\tau)^{a+1}}{(a+1)(T-\tau)}\left(a-A_1\right)^{\gamma-1}\left(A_2-a\right)^{\beta-1}da\right)^2.\end{aligned} \tag{68}$$

and

$$\begin{aligned}\mathcal{D}_{\mathcal{A}}(\tau) = {}& \frac{1}{(T-\tau)^2}\int_0^{T-\tau}\int_0^{T-\tau}\left(\left(t_1+\tau\right)^{A_1}{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log\left(t_1+\tau\right)\right)-t_1^{A_1}{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log\left(t_1\right)\right)\right)\\ &\left(\left(t_2+\tau\right)^{A_1}{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log\left(t_2+\tau\right)\right)-t_2^{A_1}{}_1F_1\left(\gamma,\gamma+\beta,\left(A_2-A_1\right)\log\left(t_2\right)\right)\right)dt_1dt_2.\end{aligned} \tag{69}$$

The expression for the first hitting time in barrier b is represented by

$$f_{\tau_b}(t) = \frac{b}{\sqrt{2\pi}\left(A_2-A_1\right)^{\gamma+\beta-1}\mathbb{B}(\gamma,\beta)}\int_{A_1}^{A_2}a\left(a-A_1\right)^{\gamma-1}\left(A_2-a\right)^{\beta-1}\exp\left\{-\frac{b^2}{2t^a}\right\}t^{-1-\frac{a}{2}}da. \tag{70}$$

To obtain the asymptotics of (70) for $t \gg 1$ let us note that $\exp\left\{-\frac{b^2}{2t^a}\right\} \in \mathcal{O}(1)$ as a function of t, a is bounded by A_2, and $t^{-1-\frac{a}{2}} \in \mathcal{O}\left(t^{-1-\frac{A_1}{2}}\right)$ for all a. Then we calculate the integral $\int_{A_1}^{A_2}\left(a-A_1\right)^{\gamma-1}\left(A_2-a\right)^{\beta-1}da$, which is equal to a constant independent on t. Finally, for (70) we have the following asymptotic formula:

$$f_{\tau_b}(t) \sim t^{-1-A_1/2}. \tag{71}$$

6. Numerical analysis

In this section, we present the results of the numerical analysis demonstrating main properties of the SBMRE discussed theoretically in the previous section for the two-point distribution. In addition, here we compare the main characteristics of SBMRE with the corresponding properties of SBM. The analysis for the case with the beta distributed anomalous diffusion exponent gives qualitatively very similar results and therefore is presented in Appendix B.

In the numerical analysis we assume $A_1 = 0.5$ and $A_2 = 1.5$. These parameters are selected to cover both subdiffusive and superdiffusive cases. Additionally, for comparison purposes, we select the same values of α in the SBM case, namely $\alpha = 0.5$ and $\alpha = 1.5$. For the two-point distribution of the anomalous diffusive exponent, we consider three p values, namely $p = 0.1$, $p = 0.5$, and $p = 0.9$. In such cases, we observe the dominance of A_2 (for $p = 0.1$), the balance between A_1 and A_2 (for $p = 0.5$), and the dominance of A_1 (for $p = 0.9$), respectively. In order to enforce similar behavior for the beta distribution of the anomalous diffusive exponent, we consider three scenarios: $(\gamma,\beta) = (0.7, 0.3)$, $(\gamma,\beta) = (0.5, 0.5)$, and $(\gamma,\beta) = (0.3, 0.7)$.

In Fig. 1 we present the exemplary trajectories for SBM for two considered α values, while in Fig. 2 we demonstrate the trajectories of SBMRE for two-point and beta distributed anomalous diffusive exponent.

It is evident that basing solely on these plots makes it difficult to distinguish between the specific processes, thus prompting a deeper exploration of their properties.

In Fig. 3, top panels, we present the pdfs depicting the position of SBMRE with the two-point distribution of the anomalous diffusion exponent for $t = 0.5$, alongside with the pdfs representing the position of subdiffusive and superdiffusive SBM. It is observed that for $p = 0.1$ the pdf of SBMRE aligns more closely with that of superdiffusive SBM, and as p increases, it tends towards the subdiffusive counterpart, consistent with intuition. In Fig. 3, bottom panels, corresponding pdfs for $t = 10$ are presented, where a

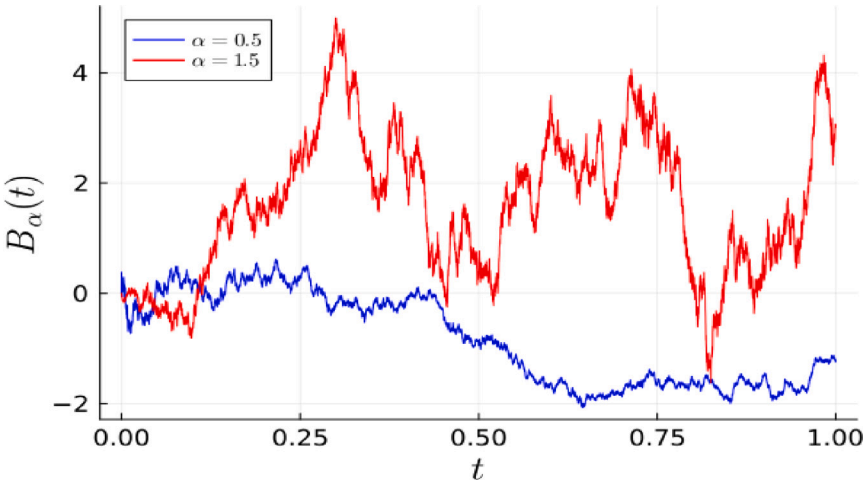

Fig. 1. Exemplary trajectories of SBM with $\alpha = 0.5$ and $\alpha = 1.5$.

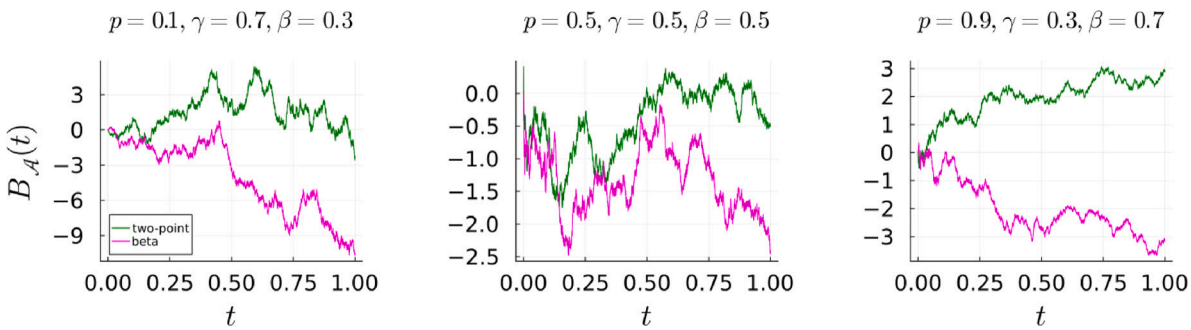

Fig. 2. Exemplary trajectories of SBMRE with a two-point and beta distribution of the anomalous diffusion exponent with different values of p, γ and β; $A_1 = 0.5$, $A_2 = 1.5$.

transition from subdiffusive to superdiffusive cases is apparent. In Fig. B.8 we demonstrate the similar plots for SBMRE with beta distributed anomalous diffusion exponent.

In Fig. 4, top panels, we demonstrate short-time behavior of the second moment of SBMRE with a two-point distribution of the anomalous diffusion exponent exhibiting characteristics akin to both subdiffusive and superdiffusive SBM (depending on p parameter). Notably, we discern that at $p = 0.1$, the SBMRE aligns more closely with the superdiffusive scenario of SBM. As the parameter p increases, there is a noticeable shift towards proximity with the subdiffusive scenario. In Fig. 4, bottom panels, a similar trend is observed for long time periods, but notably, the transition is from subdiffusive to superdiffusive SBM cases. The second moment for SBMRE with the beta-distributed exponent and SBM is presented in Fig. B.9. One can observe similar behavior as in case of the two-point distribution.

In Fig. B.10 (top panels) we depict the asymptotic formula (52) of the second moment for SBMRE for two-point distribution case for short times together with the exact formula presented in (51). Additionally, in Fig. B.10 (bottom panels) we depict the asymptotic formula (52) of the second moment for SBMRE for long times together with the exact formula presented in (51). Similar comparison for beta distribution of the anomalous diffusion exponent is presented in Fig. B.11. As can be observed, in all cases the theoretical asymptotics perfectly agree with the numerical analysis.

The expected value of TAMSD for two-point distribution of the anomalous diffusion exponent is presented in Fig. 5. Moreover, in Fig. B.12, we depict the asymptotic formula (55) of the expected value of the TAMSD for SBMRE with a two-point distribution of the anomalous diffusion exponent together with the exact value presented in (53). Here we can see the perfect agreement between theoretical formula and values received by numerical analysis. The expected value of TAMSD for the beta distributed anomalous diffusion exponent is visible in Fig. B.13. The behavior of this statistic for beta distribution is similar to the one observed for two-point distribution for appropriate cases of parameter values.

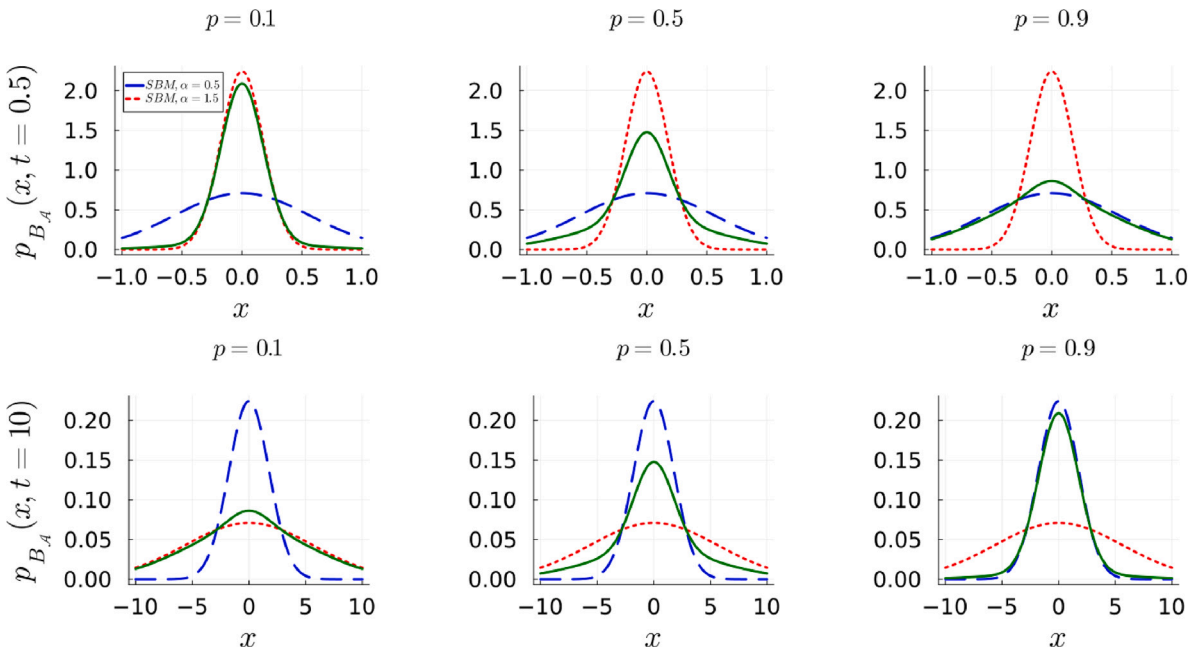

Fig. 3. Comparison of pdfs of the position of SBMRE with two-point distribution of the anomalous diffusion exponent and SBM. Top panels: $t = 0.5$. Bottom panels: $t = 10$.

Next, in Fig. 6 we plot pdfs of the first hitting time to the barrier $b = 1$ for SBMRE with the two-point distribution and compare the pdf shapes with those for both subdiffusive and superdiffusive SBM. Domination of the lower exponent at long times is clearly visible. For beta distributed anomalous diffusion exponent the same comparison is presented in Fig. B.14.

To demonstrate the interesting behavior of the EB parameter of SBMRE in case of the two-point distribution discussed in Example 5.1 in Section 5, in Fig. 7, we present the EB parameter calculated via Monte Carlo simulations and compare it with the asymptotic formula (56). The method of simulation of SBMRE is the following. First we choose time step Δt, the time horizon $T = N\Delta t$, and fix the time points $0, \Delta t, 2\Delta t, \ldots, T$. Then, we simulate the random variable $\mathcal{A}$ and rise all time point to the power $\mathcal{A}$. Finally, we simulate standard Brownian motion in such random time points. We conducted 5000 Monte Carlo simulations of the SBMRE with a two-point distributed anomalous diffusion exponent with parameters $A_1 = 0.3$, $A_2 = 0.7$, and $p = 0.5$ (note that the chosen parameters are different from those used in the first part of the presented numerical analyses due to the fact that the case of $A_1 = 0.5$ seems less interesting in the context of the EB asymptotics). The simulations were conducted for different values of τ and T. It is noteworthy that there is a very good agreement between the EB obtained from the simulations and its asymptotic behavior at $\tau/T \ll 1$.

7. Conclusions

In this paper, we investigate the probabilistic properties of SBMRE, scaled Brownian motion with random anomalous diffusion exponent. This process maintains the characteristics of SBM at the individual trajectory level, albeit possesses randomly varying exponent across the trajectories. SBM is recognized as one of the generic random processes with anomalous diffusion behavior, which finds various applications. The motivation for our study comes from modern biophysical experiments that clearly demonstrate that standard anomalous diffusion processes with fixed parameters are insufficient to describe the observed phenomena. Consequently, a natural modification of SBM is to incorporate random parameters into the "standard" anomalous diffusion processes. This scenario has been previously discussed in the context of fractional Brownian motion, leading to FBM with a random Hurst exponent [56], and more recently in the doubly stochastic version of continuous-time random walks [74] and Lévy-walk-like Langevin dynamics with random parameters [75].

Here, we expand upon the methodology of anomalous diffusion processes with random parameters by analyzing SBMRE. We establish the mathematical framework of this process by examining its pdf and the q-th absolute moment for the general distribution of the anomalous diffusion exponent. Additionally, we present the expected value of the TAMSD of SBMRE and discuss its ergodicity breaking property. Furthermore, we explore the properties of the first hitting time pdf and prove that SBMRE is a martingale, similar to its stochastic exponential.

As special cases for the distributions of the anomalous diffusion exponent, we consider the two-point and beta distributions. In these instances, we discuss the asymptotics of the characteristics of SBMRE. Our theoretical findings are validated through numerical

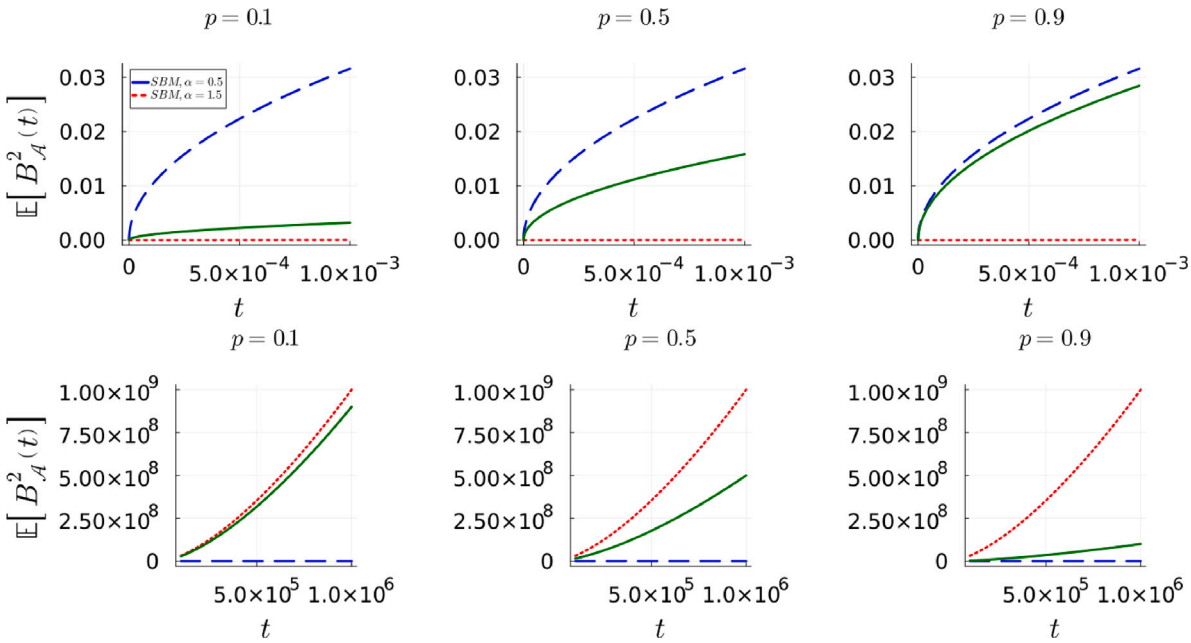

Fig. 4. Comparison of the second moments of SBMRE with two-point distribution of the anomalous diffusion exponent and SBM. Top panels: short times Bottom panels: long times.

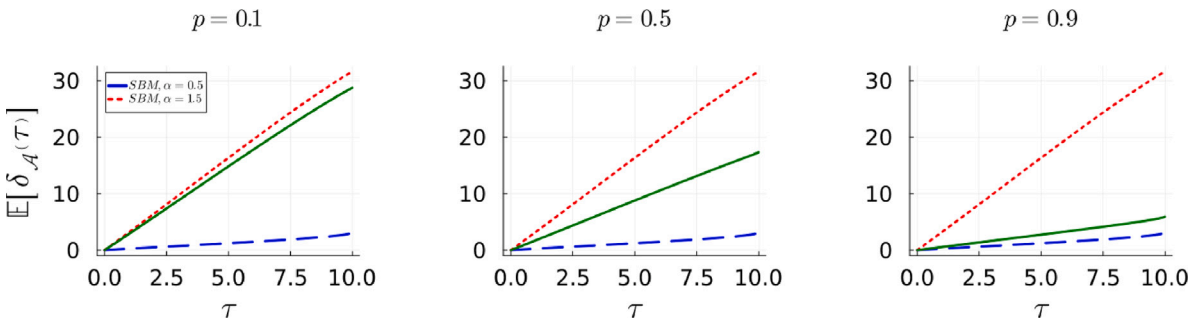

Fig. 5. Comparison of the expected values of TAMSD of SBMRE with two-point distribution of the anomalous diffusion exponent and SBM. Here $T = 10$.

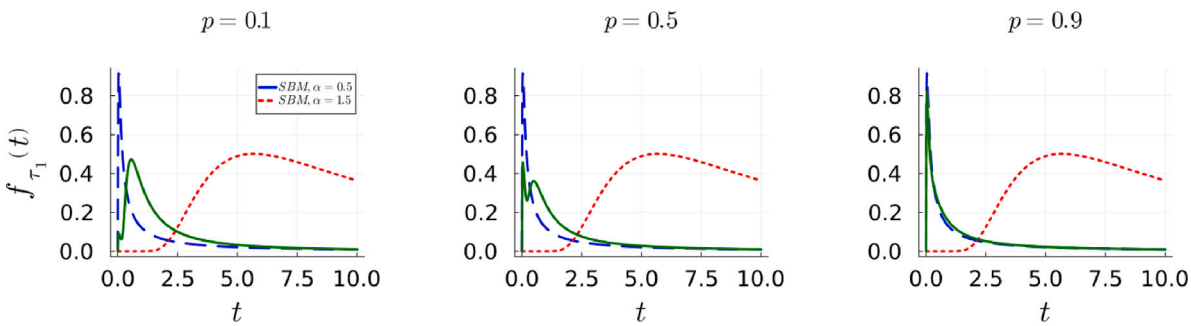

Fig. 6. Comparison of pdf of the hitting time in barrier $b = 1$ for SBMRE with two-point distribution of the anomalous diffusion exponent and SBM.

analysis and compared with similar properties of SBM. Like FBMRE, the diffusion of SBMRE is governed by the smaller exponent at short times, while for sufficiently large times, the larger exponent is dominant, an effect referred to as accelerating diffusion. Regarding the expected value of TAMSD, which is routinely measured in experiments, the dependence on trajectory length is determined by the larger exponent if the trajectory is long enough. We also observe that the long-time asymptotics of the first

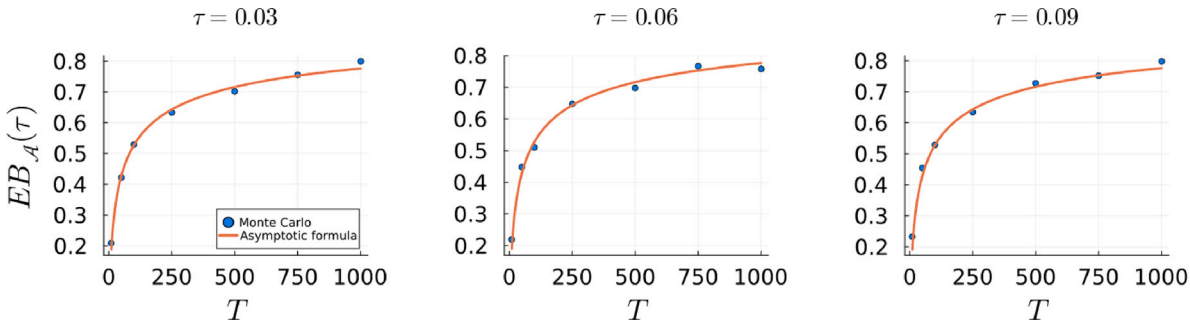

Fig. 7. EB parameter calculated using 5000 Monte Carlo simulated trajectories of SBMRE with two-point distribution of anomalous diffusion exponent compared with the asymptotic formula (56) for different values of τ and T; $A_1 = 0.3$, $A_2 = 0.7$, $p = 0.5$.

hitting time pdf of SBMRE is determined by the smaller exponent. Additionally, for the two-point distribution of the anomalous diffusion exponent, the ergodicity breaking parameter for SBMRE does not depend on the sliding window width in the limit of long trajectories, contrasting with the SBM case. A more detailed analysis of this interesting phenomenon and its extension to more realistic distributions will be provided in a forthcoming publication.

It is known that characterizing anomalous diffusion models based on measurements of individual trajectories is a challenging task that requires new approaches, particularly those based on machine learning or Bayesian inference [76]. Recently, Bayesian inference was implemented for SBM in [77] and combined with the implementation for FBM presented in [78] to demonstrate model selection and inference of the anomalous diffusion exponent for these two models. Additionally, a statistical-based approach for identifying such models was proposed in [45]. It would be interesting to extend these methods for the analysis of anomalous diffusion models with random parameters, such as SBMRE and FBMRE.

CRediT authorship contribution statement

Hubert Woszczek: Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Aleksei Chechkin:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization, Funding acquisition. **Agnieszka Wyłomańska:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Derivation of asymptotics for EB parameter for two-point distribution of anomalous diffusion exponent

Here we derive the asymptotic formulas for EB parameter for two-point distribution of anomalous diffusion exponent (36), when $\tau/T \ll 1$. Separately, we calculate the asymptotics for the functions given in (A.2), (A.3), and (A.4). Let us note that

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau) \mid \mathcal{A}\right)\right] = pEB_{A_1}(\tau)\,\mathbb{E}\left[\delta_{A_1}(\tau)\right]^2 + (1-p)\,EB_{A_2}(\tau)\,\mathbb{E}\left[\delta_{A_2}(\tau)\right]^2, \tag{A.1}$$

where $EB_{A_1}(\tau)$, $EB_{A_2}(\tau)$ are the EB parameters for SBM with $\alpha = A_1$, $\alpha = A_2$ respectively, and $\mathbb{E}\left[\delta_{A_1}(\tau)\right]^2$, $\mathbb{E}\left[\delta_{A_2}(\tau)\right]^2$ are squares of expectations of TAMSD for SBM with $\alpha = A_1$, $\alpha = A_2$ respectively.

From (37) it follows that

$$\begin{aligned}\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]=p\frac{4\tau^{2A_1+2}}{(T-\tau)^2}\Bigg[&\frac{(T/\tau-1)^{2A_1+1}}{2A_1+1}+\frac{(3A_1+1)(T/\tau-1)^{2A_1+2}}{2(A_1+1)^2(2A_1+1)}-\frac{2(T/\tau)^{A_1+1}(T/\tau-1)^{A_1+1}}{(A_1+1)^2}+\\&+\frac{(T/\tau)^{2A_1+2}}{2(A_1+1)(2A_1+1)}-\frac{(2A_1^2+A_1+1)}{2(A_1+1)^2(2A_1+1)}+\frac{2}{A_1+1}\int_0^{T/\tau-1}x^{A_1+1}(x+1)^{A_1}\,dx\Bigg]+\\+(1-p)\Bigg(\frac{4\tau^{2A_2+2}}{(T-\tau)^2}\Bigg[&\frac{(T/\tau-1)^{2A_2+1}}{2A_2+1}+\frac{(3A_2+1)(T/\tau-1)^{2A_2+2}}{2(A_2+1)^2(2A_2+1)}-\frac{2(T/\tau)^{A_2+1}(T/\tau-1)^{A_2+1}}{(A_2+1)^2}+\\&+\frac{(T/\tau)^{2A_2+2}}{2(A_2+1)(2A_2+1)}-\frac{(2A_2^2+A_2+1)}{2(A_2+1)^2(2A_2+1)}+\frac{2}{A_2+1}\int_0^{T/\tau-1}x^{A_2+1}(x+1)^{A_2}\,dx\Bigg]\Bigg).\end{aligned}\tag{A.2}$$

Additionally, we have

$$\begin{aligned}Var\left(\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right)=\mathbb{E}\left[\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]^2\right]-\mathbb{E}\left[\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right]^2=p\left(\frac{T^{A_1+1}-\tau^{A_1+1}-(T-\tau)^{A_1+1}}{(A_1+1)(T-\tau)}\right)^2+\\+(1-p)\left(\left(\frac{T^{A_2+1}-\tau^{A_2+1}-(T-\tau)^{A_2+1}}{(A_2+1)(T-\tau)}\right)^2\right)-\left(p\frac{T^{A_1+1}-\tau^{A_1+1}-(T-\tau)^{A_1+1}}{(A_1+1)(T-\tau)}+(1-p)\left(\frac{T^{A_2+1}-\tau^{A_2+1}-(T-\tau)^{A_2+1}}{(A_2+1)(T-\tau)}\right)\right)^2.\end{aligned}\tag{A.3}$$

Finally,

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right]^2=\left(p\frac{T^{A_1+1}-\tau^{A_1+1}-(T-\tau)^{A_1+1}}{(A_1+1)(T-\tau)}+(1-p)\left(\frac{T^{A_2+1}-\tau^{A_2+1}-(T-\tau)^{A_2+1}}{(A_2+1)(T-\tau)}\right)\right)^2.\tag{A.4}$$

To obtain (A.2), (A.3), (A.4) we use formula (48) for the pdf of two-point distribution.

Let us note that the asymptotics of $\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]$ is equal to the asymptotics of the EB parameter for SBM (see (15)) multiplied by the square of the asymptotic of expectation of TAMSD for SBM, see (12). Since the asymptotic of the EB parameter is different for different ranges of parameter α we have to divide our analysis into few cases. First, we assume $A_1, A_2 < 1/2$. In this case (A.1) takes the form

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]\sim pC(A_1)\frac{\tau^{2A_1+2}}{T^2}+(1-p)C(A_2)\frac{\tau^{2A_2+2}}{T^2},\tag{A.5}$$

Further, for $A_1 < 1/2$ and $A_2 = 1/2$

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]\sim pC(A_1)\frac{\tau^{2A_1+2}}{T^2}+(1-p)\frac{\tau^3}{12T^{3-2A_2}}\left[\log\left(\frac{T}{\tau}\right)+2\log(2)-\frac{5}{6}\right],\tag{A.6}$$

for $A_1 < 1/2$ and $A_2 > 1/2$

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]\sim pC(A_1)\frac{\tau^{2A_1+2}}{T^2}+(1-p)\frac{A_2^2}{3(2A_2-1)}\frac{\tau^3}{T^{3-2A_2}},\tag{A.7}$$

and for $A_1 = 1/2$ and $A_2 > 1/2$

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]\sim p\frac{\tau^3}{12T^{3-2A_1}}\left[\log\left(\frac{T}{\tau}\right)+2\log(2)-\frac{5}{6}\right]+(1-p)\frac{A_2^2}{3(2A_2-1)}\frac{\tau^3}{T^{3-2A_2}}.\tag{A.8}$$

Finally, for $A_1, A_2 > 1/2$ the following holds:

$$\mathbb{E}\left[Var\left(\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right)\right]\sim p\frac{A_2^2}{3(2A_2-1)}\frac{\tau^3}{T^{3-2A_1}}+(1-p)\frac{A_2^2}{3(2A_2-1)}\frac{\tau^3}{T^{3-2A_2}}.\tag{A.9}$$

For $Var\left(\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right)$ we have the following asymptotics when $\tau/T \ll 1$:

$$\begin{aligned}Var\left(\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\mid\mathcal{A}\right]\right)\sim\left[\left(\frac{\tau}{T}\right)^2\left(pT^{2A_1}+(1-p)T^{2A_2}\right)\right]-\left(\frac{\tau}{T}\left(pT^{A_1}+(1-p)T^{A_2}\right)\right)^2=\\=\left(\frac{\tau}{T}\right)^2\left[\left(p-p^2\right)\left(T^{2A_1}+T^{2A_2}\right)-2p(1-p)T^{A_1+A_2}\right].\end{aligned}\tag{A.10}$$

The above is the straightforward consequence of the asymptotics of expectation of TAMSD for SBM presented in (11). Let us note, in case $A_1 = A_2 = 1/2$ the SBMRE reduces to SBM with $\alpha = 1/2$ and thus this case is not considered here. Finally, for (A.4) the following holds, when $\tau/T \ll 1$

$$\mathbb{E}\left[\delta_{\mathcal{A}}(\tau)\right]^2\sim\left(\frac{\tau}{T}\left(pT^{A_1}+(1-p)T^{A_2}\right)\right)^2,\tag{A.11}$$

which is just the square of the asymptotics of expectation of TAMSD for SBMRE with two-point distribution of anomalous diffusion exponent, see (55). Taking formulas (A.5)–(A.11) we obtain the asymptotics for the EB parameter for two-point distributed anomalous diffusion exponent given by (56).

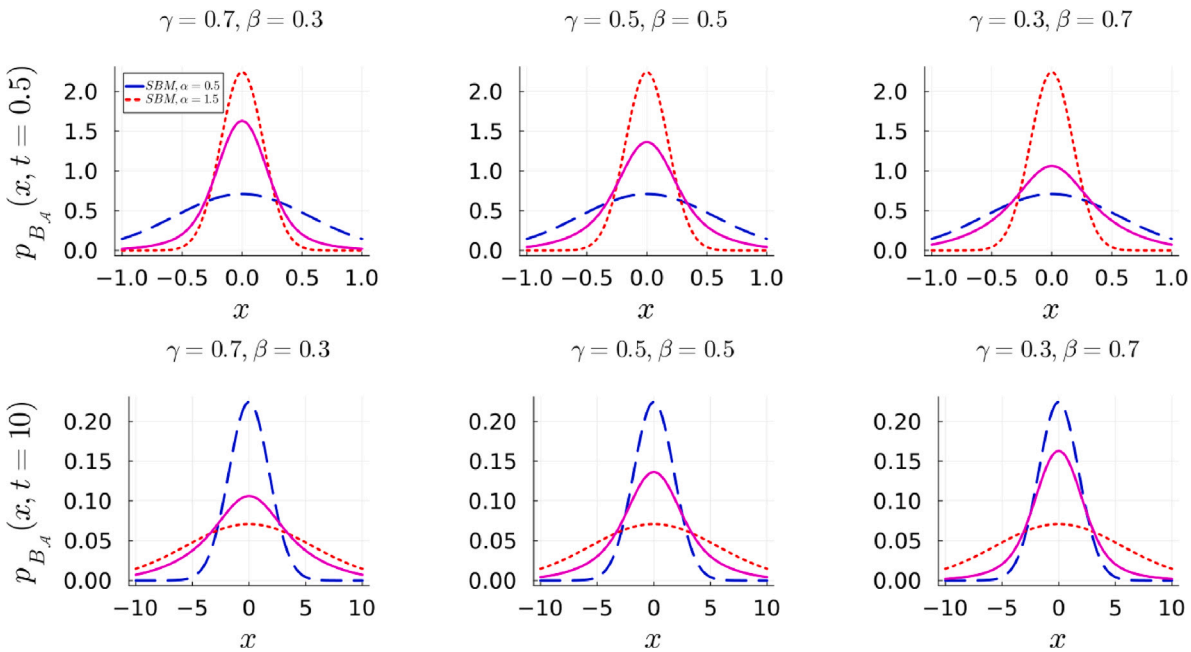

Fig. B.8. Comparison of pdfs of the position of SBMRE with beta distribution of the anomalous diffusion exponent and SBM. Top panels: $t = 0.5$. Bottom panels: $t = 10$.

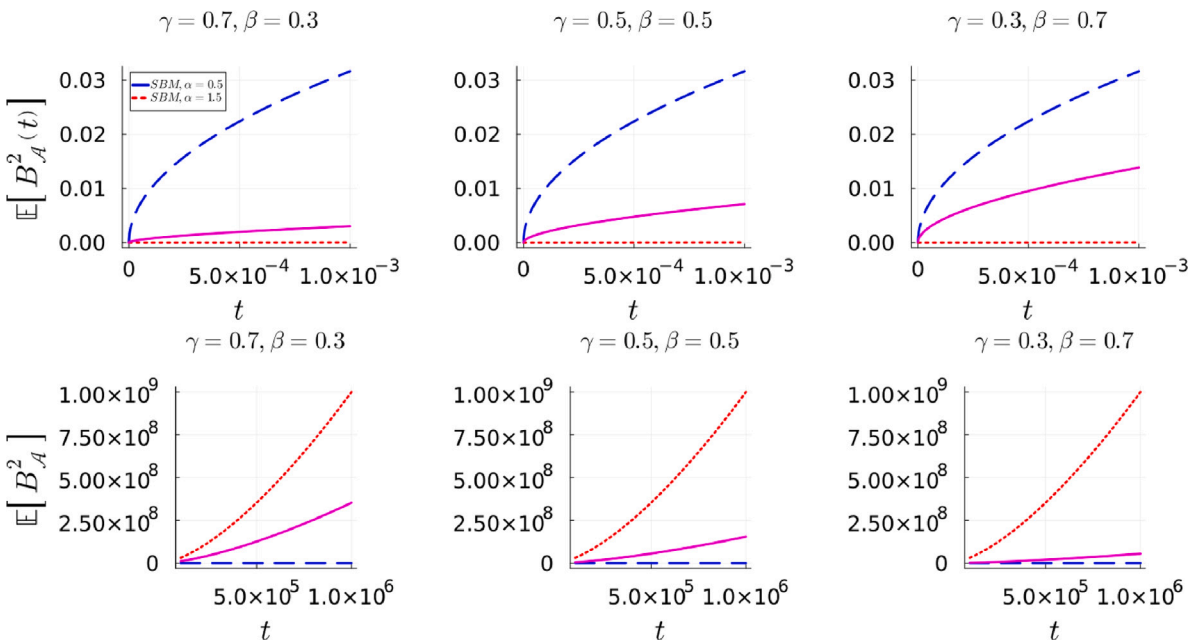

Fig. B.9. Comparison of second moments of SBMRE with beta distribution of the anomalous diffusion exponent and SBM. Top panels: short times. Bottom panels: long times.

Appendix B. Additional plots

See Figs. B.8–B.14.

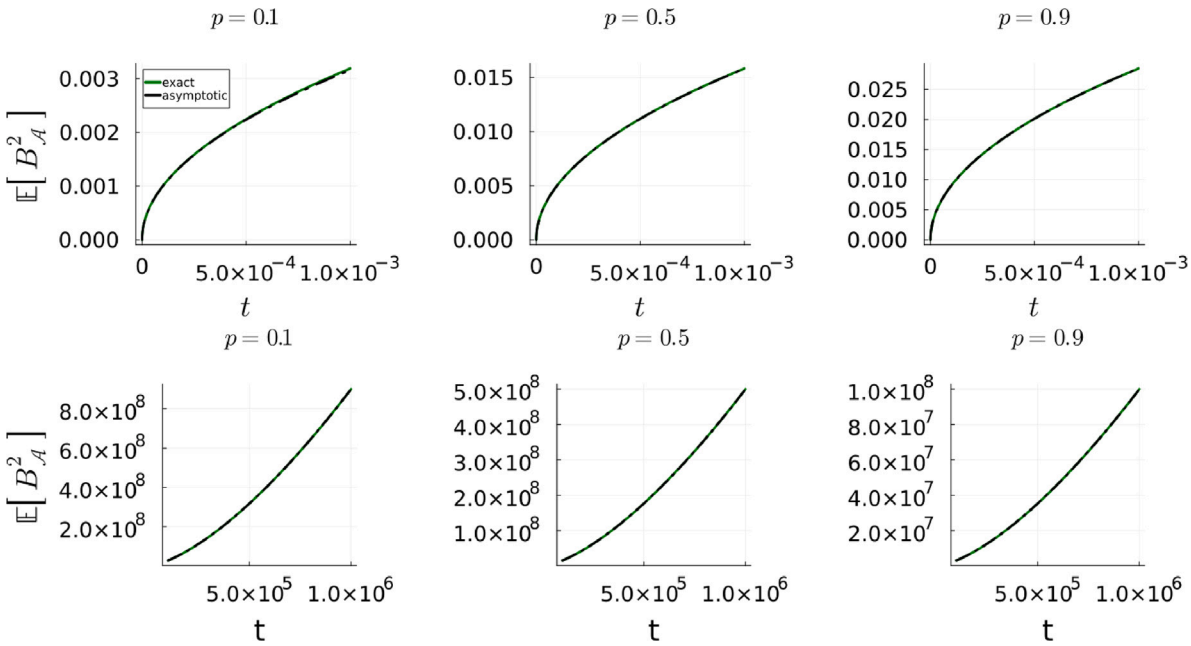

Fig. B.10. Comparison of second moments of SBMRE with two-point distribution of the anomalous diffusion exponent and their asymptotics (see formula (52)). Top panels: short times. Bottom panels: long times.

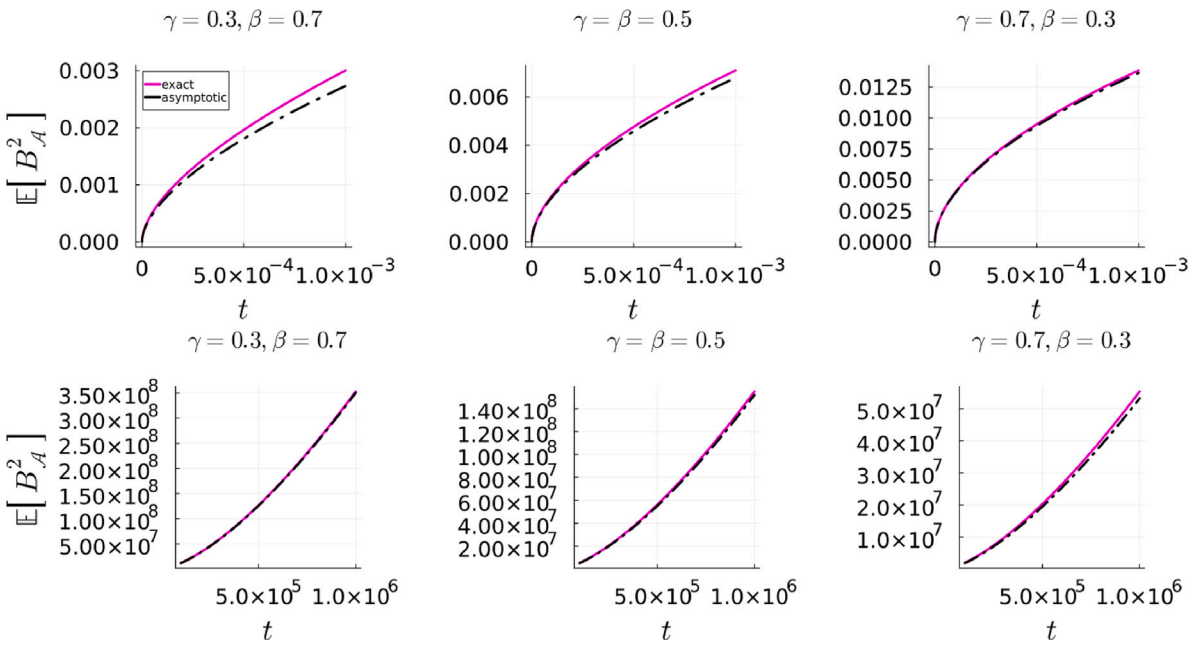

Fig. B.11. Comparison of second moments of SBMRE with beta distribution of the anomalous diffusion exponent and their asymptotics (see formulas (64), (65)). Top panels: short times. Bottom panels: long times.

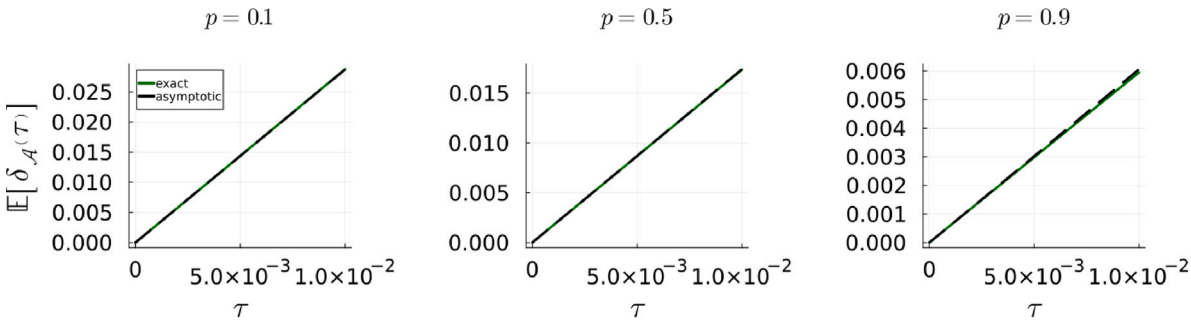

Fig. B.12. Asymptotic (see formula (55)) of the expectation of the TAMSD of SBMRE with a two-point distribution of the anomalous diffusion exponent. Here $T = 10$.

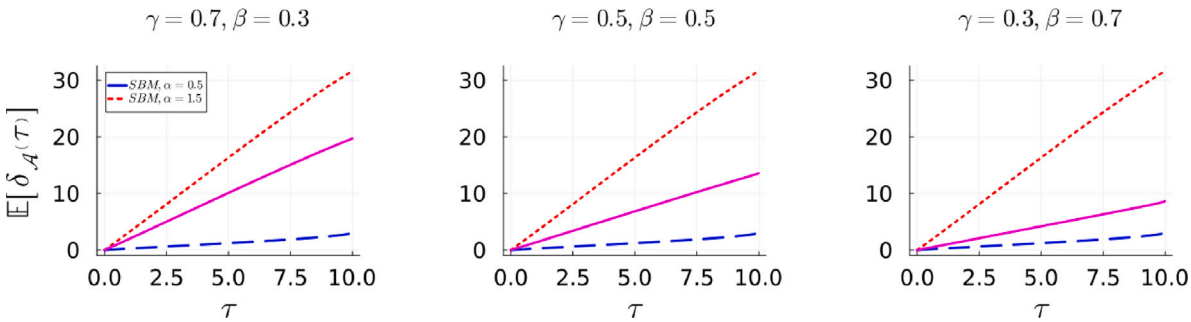

Fig. B.13. Comparison of the expected values of TAMSD of SBMRE with beta distribution of the anomalous diffusion exponent and SBM. Here $T = 10$.

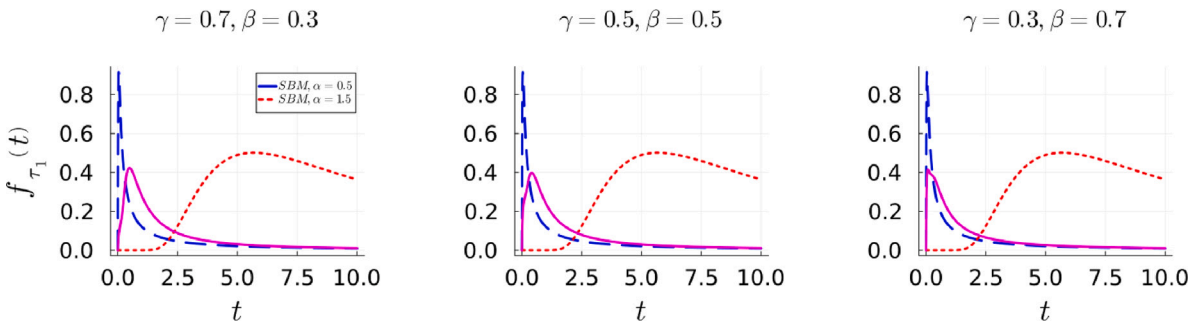

Fig. B.14. Comparison of pdf of the hitting time in barrier $b = 1$ for SBMRE with beta distribution of the anomalous diffusion exponent and SBM.

Data availability

No data was used for the research described in the article.

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